Lecture 1

Linear representations and complete reducibility

Objectives. *Symmetry enters physics through linear spaces: quantum states superpose, small oscillations add. This lecture sets up the central notion of the course — a linear representation of a group — together with the construction by which symmetries act on functions, and proves the two structural theorems that make finite group representations tractable: every representation can be made unitary by averaging, and every representation decomposes into irreducible pieces (Maschke's theorem). The permutation representation of S_3 is decomposed in full as a model computation.*

1.1 Representations: definition and first examples

Lectures 1 and 2 studied groups in the abstract. Physics, however, meets a group through its *action on the states of a system*, and the state spaces of physics — quantum Hilbert spaces, spaces of displacements of atoms, spaces of fields — are vector spaces. The operations of the group must respect that linear structure: a symmetry operation applied to a superposition is the superposition of the transformed states. The mathematical object capturing this is the following.

Definition 1.1 (Representation)**.** A *(linear) representation* of a group G on a finite-dimensional complex vector space V is a homomorphism

$$\rho\colon G \longrightarrow \mathrm{GL}(V),$$

where $\mathrm{GL}(V)$ is the group of invertible linear operators on V. Thus to each $g \in G$ is assigned an invertible operator $\rho(g)$, with

$$\rho(gh) = \rho(g)\,\rho(h) \quad \text{for all } g, h \in G,$$

whence automatically $\rho(e) = \mathrm{id}_V$ and $\rho(g^{-1}) = \rho(g)^{-1}$. The *dimension* (or degree) of the representation is $\dim V$. We speak of "the representation (ρ, V)", or simply of V when ρ is understood.

Unless stated otherwise our vector spaces are complex and finite-dimensional; the real case will be flagged when it matters (and it will matter — see Example 1.13).

Remark 1.2 (Scope for this part of the course)**.** This standing definition is the first rung of the representation-scope ladder in the student syllabus: algebraic finite-dimensional representations. Later in this lecture we also construct representations on function spaces, which may be infinite-dimensional, but Maschke's theorem, finite character sums, trace identities, and dimension counts are used only under their stated finite-group and finite-dimensional hypotheses. Part II separately adds continuity, unitarity, projective actions, and domains of unbounded derived generators; those are not hidden in Definition 1.1.

Notation: operators versus matrices. Throughout these notes we keep two parallel notations, and the distinction is deliberate. The symbol $\rho(g)$ always denotes the abstract linear *operator* on V, defined without reference to any basis. Once a basis $(e_1, \dots, e_n)$ of V is chosen, the corresponding *matrices* are denoted $D(g)$ and defined by

$$\rho(g)\, e_j = \sum_{i=1}^{n} D_{ij}(g)\, e_i,$$

so that the j-th column of $D(g)$ lists the components of the image of the j-th basis vector. The homomorphism property becomes matrix multiplication, $D(gh) = D(g)D(h)$, and a map $D\colon G \to \mathrm{GL}(n, \mathbb{C})$ with this property is called a *matrix representation*. Irreducible representations, once we have them, will carry labels $\rho^{(i)}$, $D^{(i)}$. Keeping operators and matrices typographically apart costs little and prevents a great deal of confusion when several bases are in play at once — as they constantly are from Section 1.6 onwards.

Example 1.3 (One-dimensional representations)**.** On a one-dimensional space, an invertible operator is multiplication by a nonzero number: $\mathrm{GL}(1, \mathbb{C}) = \mathbb{C} \setminus \{0\}$. A one-dimensional representation is therefore nothing but a homomorphism $G \to \mathbb{C} \setminus \{0\}$, and we have already met several:

(i) the *trivial representation* $\rho(g) = 1$ for all g, defined for every group, on which symmetry acts invisibly;
(ii) the *sign representation* $\mathrm{sgn}\colon S_n \to \{\pm 1\}$ of Lecture 1;
(iii) the *orientation representation* $\nu\colon D_n \to \{\pm 1\}$ of Lecture 2, with $\nu = +1$ on rotations and -1 on reflections.

Example 1.4 (All the χ_k of a cyclic group)**.** Let $C_n = \langle r \rangle$ and $\omega = e^{2\pi \mathrm{i}/n}$. For each $k \in \{0, 1, \dots, n-1\}$,

$$\chi_k(r^m) = \omega^{km} = e^{2\pi \mathrm{i} km/n}$$

is a one-dimensional representation: $\chi_k(r^a)\chi_k(r^b) = \omega^{k(a+b)} = \chi_k(r^{a+b})$, and the value is well defined because $\omega^{kn} = 1$. These n representations are pairwise distinct ($\chi_k(r) = \omega^k$ differ), and in Lecture 4 we shall see that they are *all* the irreducible representations of C_n — the finite Fourier modes, as we will recognise when characters appear.

Example 1.5 (The defining representation of D_n)**.** Realising D_n by orthogonal transformations of the plane, as in Lecture 1, *is* a two-dimensional real representation, called the *defining* (or vector) representation. In the basis of Sheet 1 (vertex 1 on the y-axis), the generators of D_3 have matrices

$$D(r) = \begin{pmatrix} \cos\frac{2\pi}{3} & -\sin\frac{2\pi}{3} \\ \sin\frac{2\pi}{3} & \cos\frac{2\pi}{3} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

and one checks $D(r)^3 = D(s)^2 = \mathbb{1}$, $D(s)D(r)D(s)^{-1} = D(r)^{-1}$: the dihedral relations hold, so the assignment extends consistently to all of D_3.

Groups often arrive acting on a *set* before they act on a vector space; the following definition names what we have in fact been doing since Lecture 1.

Definition 1.6 (Group action)**.** An *action* of a group G on a set X is a map $G \times X \to X$, written $(g, x) \mapsto g \cdot x$, such that $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$, $x \in X$. Symmetry operations acting on the points of space, S_n acting on $\{1, \dots, n\}$, and G acting on itself by conjugation (Lecture 2) are all actions.

Example 1.7 (Permutation representations)**.** Let G act on a finite set X. Take a vector space with one basis vector e_x for each point $x \in X$, namely $V = \mathbb{C}^X$, and let G permute the basis the way it permutes the points:

$$\rho(g)\, e_x = e_{g \cdot x}.$$

Then $\rho(g)\rho(h)e_x = e_{g\cdot(h\cdot x)} = e_{(gh)\cdot x} = \rho(gh)e_x$: a representation of dimension $|X|$, the *permutation representation* of the action. Its matrices are *permutation matrices* — a single 1 in each row and column. For S_3 acting on $\{1, 2, 3\}$, the generators $(1\,2)$ and $(1\,2\,3)$ act on the basis (e_1, e_2, e_3) by

$$D\big((1\,2)\big) = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad D\big((1\,2\,3)\big) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

the columns recording $e_1 \mapsto e_2$, $e_2 \mapsto e_3$, $e_3 \mapsto e_1$ in the second case. This representation is dissected completely in Section 1.7. Taking $X = G$ with the action of G on itself by left multiplication yields the *regular representation*, of dimension $|G|$, whose remarkable properties drive Lecture 4.

Remark 1.8 (Faithfulness)**.** A representation is *faithful* if ρ is injective ($\ker \rho = \{e\}$), i.e. if the matrices remember the whole group. The defining representation of D_n and all permutation representations of group actions that separate points are faithful; the trivial representation is maximally unfaithful. There is no requirement of faithfulness in the theory: a representation may legitimately see only a quotient $G/\ker \rho$ — recall the orientation representation, which sees only orientation.

1.2 How symmetries act on functions

We now come to the single most important construction of the course. The states of physical systems are very often *functions* on the space the symmetry group transforms: wavefunctions on configuration space, displacement fields on a molecule, temperature distributions on a body. The group's action on points then induces an action on the functions — and it does so by a formula whose one subtle feature, an inverse, repays careful attention.

Theorem 1.9 (The function-space representation)**.** *Let the group G act on a set X, and let $\mathcal{F}(X)$ denote the vector space of complex-valued functions on X. For $g \in G$ and $\psi \in \mathcal{F}(X)$ define*

$$\big(\rho(g)\psi\big)(x) := \psi\big(g^{-1} \cdot x\big).$$

Then each $\rho(g)$ is an invertible linear operator and $\rho\colon G \to \mathrm{GL}(\mathcal{F}(X))$ is a representation. The same formula defines a representation on any subspace of $\mathcal{F}(X)$ stable under all the $\rho(g)$ (for infinite X one works with such subspaces — polynomials, smooth functions, square-integrable functions — as the context requires).

Proof. Linearity in ψ is clear, and $\rho(e) = \mathrm{id}$. The homomorphism property is a computation worth doing once and remembering forever:

$$\big(\rho(g)\rho(h)\psi\big)(x) = \big(\rho(h)\psi\big)(g^{-1} \cdot x) = \psi\big(h^{-1} \cdot (g^{-1} \cdot x)\big) = \psi\big((gh)^{-1} \cdot x\big) = \big(\rho(gh)\psi\big)(x),$$

using $h^{-1}g^{-1} = (gh)^{-1}$ from Lecture 1. Invertibility follows: $\rho(g)\rho(g^{-1}) = \rho(e) = \mathrm{id}$. ■

Why the inverse? First the conceptual answer: $\rho(g)\psi$ is the function ψ *carried along by* g. Evaluating the definition at a transformed point,

$$\big(\rho(g)\psi\big)(g \cdot x) = \psi(x) :$$

the transported function takes, at the transported point, the value the original function took at the original point — rotate a temperature distribution by g and the hot spot moves *with* the rotation. Second, the mechanical answer: the inverse is forced by the order of composition. Had we defined $(\sigma(g)\psi)(x) = \psi(g \cdot x)$, the same computation would give $\sigma(g)\sigma(h) = \sigma(hg)$ — the products in the wrong order, an *anti*-homomorphism. The inverse undoes the reversal. Lecture 1's polynomial action is the same mechanism written in coordinate slots. With the usual left action $\sigma \cdot (x_1, \ldots, x_n) = (x_{\sigma^{-1}(1)}, \ldots, x_{\sigma^{-1}(n)})$, its inverse sends x to $(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$, so

$$(P_\sigma f)(x_1, \ldots, x_n) = f(x_{\sigma(1)}, \ldots, x_{\sigma(n)}).$$

Thus Lecture 1's direct coordinate substitution is precisely this valid inverse pullback; confusing the action on coordinate slots with its pullback is what reverses the product.

Example 1.10 (Coordinate functions; the defining representation recovered)**.** Let $G \subseteq \mathrm{O}(2)$ act on the plane, and consider the two-dimensional space of *linear* functions spanned by the coordinate functions x_1, x_2 (so $x_i(\vec{r}\,) = r_i$). This space is stable:

$$\big(\rho(g)x_i\big)(\vec{r}\,) = x_i\big(g^{-1}\vec{r}\,\big) = (g^{-1})_{ij}\, r_j = (g^{\mathsf{T}})_{ij}\, r_j = g_{ji}\, x_j(\vec{r}\,),$$

summation over j understood and orthogonality $g^{-1} = g^{\mathsf{T}}$ used. Hence $\rho(g)x_i = \sum_j g_{ji}x_j$: comparing with the convention $\rho(g)e_j = \sum_i D_{ij}(g)e_i$, the matrix of $\rho(g)$ in the basis (x_1, x_2) is exactly g itself. For $G = D_3$ the function-space representation on linear functions *reproduces the defining representation* of Example 1.5. Higher-degree polynomials furnish a tower of further representations — the quadratics x_1^2, x_1x_2, x_2^2 carry a three-dimensional one — whose decomposition into irreducibles is a short computation once we have the tools of Lecture 5.

1.2.1 The dual representation

For a general (not necessarily orthogonal) representation, the computation above does not return the same matrices, and what it does return deserves its own name. Recall that the *dual space* V^* of V is the space of linear maps $\varphi\colon V \to \mathbb{C}$ (*functionals*); given a basis (e_i) of V, the dual basis (e^i) of V^* is defined by $e^i(e_j) = \delta_{ij}$. Linear functionals are functions on V, so the function-space construction applies to them, and it preserves linearity:

Proposition 1.11 (Dual representation)**.** *Let (ρ, V) be a representation. Then*

$$\big(\rho^*(g)\varphi\big)(v) := \varphi\big(\rho(g^{-1})v\big), \qquad \varphi \in V^*,\ v \in V,$$

defines a representation (ρ^*, V^*), the dual *(or* contragredient*) representation. In the dual basis its matrices are*

$$D^*(g) = D(g^{-1})^{\mathsf{T}} = \big(D(g)^{-1}\big)^{\mathsf{T}}.$$

Proof. Each $\rho^*(g)$ maps functionals to functionals linearly, and the homomorphism property is Theorem 1.9 applied to the action of G on the set V via ρ. For the matrices: $\big(\rho^*(g)e^j\big)(e_i) =$

$e^j\big(\rho(g^{-1})e_i\big) = D_{ji}(g^{-1})$, so expanding $\rho^*(g)e^j$ in the dual basis gives coefficients $D^*_{ij}(g) = D_{ji}(g^{-1})$, which is the stated matrix. One can also verify directly that $g \mapsto D(g^{-1})^{\mathsf{T}}$ is a homomorphism: transposition and inversion each reverse the order of products, and two reversals cancel. ■

If the matrices $D(g)$ are *unitary* — and the next section shows this can always be arranged — then $D(g^{-1}) = D(g)^\dagger$ and $D^*(g) = \overline{D(g)}$: *the dual representation is the complex conjugate representation.* The physics reader has met this contragredient behaviour before: if kets transform as $|\psi\rangle \mapsto \rho(g)\,|\psi\rangle$, then bras transform with the conjugate matrices so that $\langle\varphi|\psi\rangle$ stays invariant. The distinction between a representation and its conjugate, invisible when the matrices are real, prepares an important distinction in later SU(3) study: the quark triplet **3** and the antiquark antitriplet $\bar{\mathbf{3}}$ are conjugate representations. This course builds the dual/conjugate vocabulary but does not teach that particle-physics application; Jones, Ch. 8, §§8.3–8.4 is a further-reading destination.

The function-space construction is fully general, and we shall lean on it repeatedly. It is how a finite group acts on polynomials (the sign of a permutation, Lecture 1, and Example 1.10); how a point group acts on the displacement patterns of a vibrating molecule (Lecture 6); how a symmetry group acts on quantum wavefunctions, $\psi(\vec r\,) \mapsto \psi(g^{-1}\vec r\,)$, which is the formula underlying degeneracies and selection rules (Lecture 7); and how the rotation group acts on functions on the sphere, whose stable subspaces are spanned by the spherical harmonics (Part II). One formula, the whole course.

1.3 Equivalence of representations

When are two representations "the same"? As with groups themselves (Lecture 1), the answer is: when a structure-preserving dictionary translates one into the other.

Definition 1.12 (Equivalence)**.** Representations (ρ_V, V) and (ρ_W, W) of the same group G are *equivalent*, written $\rho_V \cong \rho_W$, if there is an invertible linear map $T : V \to W$ with

$$T\,\rho_V(g) = \rho_W(g)\,T \qquad \text{for all } g \in G.$$

Such a T is called an *intertwiner* (it lets the group action pass through it).

In matrix language: choosing bases and letting S be the matrix of T, equivalence reads $D_W(g) = S\,D_V(g)\,S^{-1}$ for all g, with one and the same S. In particular the matrices of a single representation in two different bases are related this way — equivalent representations are one representation written in different coordinates, and all notions we build (irreducibility, characters, multiplicities) will be invariant under equivalence.

Example 1.13 (A representation that splits over $\mathbb{C}$ but not over $\mathbb{R}$)**.** Let $C_3 = \langle r\rangle$ act on $\mathbb{C}^2$ through the rotation matrices of Example 1.5: $D(r^k) = R(2\pi k/3)$. The vectors $v_\pm = \binom{1}{\mp \mathrm{i}}$ are eigenvectors of every rotation:

$$R(\theta)\begin{pmatrix}1\\ \mp\mathrm{i}\end{pmatrix} = \begin{pmatrix}\cos\theta \pm \mathrm{i}\sin\theta\\ \sin\theta \mp \mathrm{i}\cos\theta\end{pmatrix} = e^{\pm\mathrm{i}\theta}\begin{pmatrix}1\\ \mp\mathrm{i}\end{pmatrix},$$

as multiplying out confirms. So in the basis (v_+, v_-) the matrices become

$$S^{-1}D(r^k)\,S = \begin{pmatrix}\omega^k & 0\\ 0 & \omega^{-k}\end{pmatrix}, \qquad \omega = e^{2\pi\mathrm{i}/3}, \quad S = (v_+\,|\,v_-)\,:$$

the rotation representation of C_3 is equivalent, over $\mathbb{C}$, to the direct sum $\chi_1 \oplus \chi_{-1}$ of two of the one-dimensional representations of Example 1.4. Notice that the diagonalising basis is genuinely complex: over the *reals* a rotation through $2\pi/3$ has no eigenvector at all, and the representation cannot be split. Working over $\mathbb{C}$ is not a convenience but a structural choice, and it is the right one for quantum mechanics, where state spaces are complex from the outset.

1.4 Invariant subspaces and irreducibility

Example 1.13 displayed a representation containing smaller ones. The general notions:

Definition 1.14 (Invariant subspace; irreducibility)**.** Let (ρ, V) be a representation. A subspace $W \subseteq V$ is *invariant* if $\rho(g)W \subseteq W$ for all $g \in G$ (equivalently $\rho(g)W = W$: apply the condition to g^{-1} as well). An invariant subspace carries the *subrepresentation* $\rho|_W : g \mapsto \rho(g)|_W$. The representation is *irreducible* if $\dim V \geq 1$ and its only invariant subspaces are $\{0\}$ and V; otherwise it is *reducible*. Every one-dimensional representation is irreducible.

Example 1.15 (Inside the permutation representation)**.** In the permutation representation of S_n on $\mathbb{C}^n$ (Example 1.7), the line

$$L = \mathbb{C}\, u_0, \qquad u_0 = e_1 + e_2 + \cdots + e_n,$$

is invariant: permuting the summands leaves u_0 fixed, so L carries a copy of the trivial representation. A second invariant subspace is

$$V_0 = \left\{ v \in \mathbb{C}^n : \textstyle\sum_i v_i = 0 \right\},$$

of dimension $n-1$: permuting the components of v does not change their sum. So the permutation representation is reducible for every $n \geq 2$, and $\mathbb{C}^n = L \oplus V_0$ as vector spaces, both summands invariant. The subrepresentation on V_0 is called the *standard representation* of S_n; for $n = 3$ we prove it irreducible in Section 1.7.

In the example, the invariant subspace came with an invariant complement. That this is automatic for finite groups is the content of Maschke's theorem — and it is genuinely a theorem, not an algebraic triviality:

Example 1.16 (Reducible but indecomposable)**.** Let the *infinite* group $(\mathbb{Z}, +)$ be represented on $\mathbb{C}^2$ by

$$\rho(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, \qquad \rho(n)\rho(m) = \begin{pmatrix} 1 & n+m \\ 0 & 1 \end{pmatrix} = \rho(n+m).$$

The line $\mathbb{C}e_1$ is invariant, so the representation is reducible. But no *complementary* invariant line exists: any line other than $\mathbb{C}e_1$ is spanned by some $\binom{a}{1}$, and $\rho(1)\binom{a}{1} = \binom{a+1}{1}$ is never proportional to $\binom{a}{1}$. The representation cannot be written as a direct sum of smaller ones — the matrices can be made simultaneously triangular but never simultaneously block-diagonal. Finiteness of the group is exactly what will rule this pathology out.

1.5 Unitarity: the averaging trick

Equip V with a Hermitian inner product $\langle \cdot, \cdot \rangle$, conjugate-linear in the first argument (the physics convention). An operator U is *unitary* if $\langle Uu, Uv \rangle = \langle u, v \rangle$ for all u, v, and a representation is *unitary* (with respect to the given product) if every $\rho(g)$ is.

Theorem 1.17 (Unitarisability)**.** *Let (ρ, V) be a representation of a* finite *group G, and let $\langle\cdot,\cdot\rangle_0$ be any inner product on V. Then*

$$\langle u, v\rangle \;:=\; \frac{1}{|G|}\sum_{h\in G}\langle \rho(h)u,\ \rho(h)v\rangle_0$$

is an inner product on V with respect to which ρ is unitary.

Proof. Sesquilinearity is inherited term by term, and positivity holds because a sum of nonnegative terms vanishes only if each term does — in particular the $h = e$ term $\langle u, u\rangle_0$, so $\langle u, u\rangle = 0$ forces $u = 0$. For unitarity, fix g and compute

$$\langle \rho(g)u, \rho(g)v\rangle = \frac{1}{|G|}\sum_{h\in G}\langle \rho(hg)u, \rho(hg)v\rangle_0 = \frac{1}{|G|}\sum_{h'\in G}\langle \rho(h')u, \rho(h')v\rangle_0 = \langle u, v\rangle\,,$$

because as h runs once over G, so does $h' = hg$: the map $h \mapsto hg$ is a bijection of G (cancellation — the Latin-square property of the Cayley table, Lecture 1). ∎

Remark 1.18 (The averaging trick)**.** The proof's engine — *average a quantity over the group, and the average is automatically invariant because translating the summation variable merely reshuffles the sum* — is the most frequently reused idea in this subject. It will prove Maschke's theorem in a moment, the orthogonality relations in Lecture 4, and the projection formulae in Lecture 5; and in Part II, with the sum replaced by an invariant integral over a compact Lie group, the entire theory will be rebuilt on it.

In matrix terms, the theorem says every matrix representation of a finite group is equivalent to one by unitary matrices: take any basis of V orthonormal for the invariant product; the change of basis S conjugates the $D(g)$ into unitary form.

Example 1.19 (Averaging, concretely)**.** Let $C_2 = \{e, a\}$ act on $\mathbb{C}^2$ by

$$D(e) = \mathbb{1}, \qquad D(a) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad D(a)^2 = \mathbb{1}\ \checkmark,$$

a representation that is *not* unitary for the standard product: $D(a)^\dagger D(a) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \neq \mathbb{1}$. Averaging the standard product gives $\langle u, v\rangle = u^\dagger M v$ with

$$M = \tfrac{1}{2}\Big(\mathbb{1} + D(a)^\dagger D(a)\Big) = \frac{1}{2}\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix},$$

and one verifies $D(a)^\dagger M D(a) = M$: the representation is unitary for the new product. Equivalently, pass to the eigenbasis: $D(a)$ has eigenvectors $v_+ = \left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ and $v_- = \left(\begin{smallmatrix} 1 \\ -2 \end{smallmatrix}\right)$ with eigenvalues ± 1, and these are orthogonal in the *invariant* product (compute: $Mv_- = \left(\begin{smallmatrix} 0 \\ -5/2 \end{smallmatrix}\right)$, so $v_+^\dagger M v_- = 0$) though not in the standard one. In the basis (v_+, v_-) the matrix becomes $\mathrm{diag}(1, -1)$ — manifestly unitary, and incidentally displaying the decomposition of this representation into the two one-dimensional representations of C_2.

The payoff of unitarity is structural:

Corollary 1.20. *In a unitary representation, the orthogonal complement of an invariant subspace is invariant.*

Proof. Let W be invariant and $w' \in W^\perp$. For any $g \in G$ and $w \in W$,

$$\langle \rho(g)w',\ w\rangle = \big\langle w',\ \rho(g)^{-1}w\big\rangle = \big\langle w',\ \rho(g^{-1})w\big\rangle = 0,$$

the first step by unitarity and the last because $\rho(g^{-1})w \in W$. Hence $\rho(g)w' \in W^\perp$. ∎

1.6 Maschke's theorem: complete reducibility

Theorem 1.21 (Maschke)**.** *Every finite-dimensional complex representation of a finite group is a direct sum of irreducible subrepresentations: there are invariant subspaces $W_1, \dots, W_k$, each irreducible, with*

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_k.$$

Proof. Induct on $\dim V$. If V is irreducible (in particular if $\dim V = 1$), there is nothing to prove. Otherwise, equip V with an invariant inner product (Theorem 1.17) and let W be an invariant subspace with $0 \neq W \neq V$. By Corollary 1.20, $V = W \oplus W^{\perp}$ with both summands invariant and of strictly smaller dimension; apply the induction hypothesis to each. ■

In matrix language: *there is a single invertible S such that*

$$S\,D(g)\,S^{-1} = \begin{pmatrix} D^{(i_1)}(g) & & \\ & D^{(i_2)}(g) & \\ & & \ddots \end{pmatrix} \qquad \textit{for every } g \in G$$

simultaneously, with each diagonal block an irreducible matrix representation (Figure 1.1). The force of the statement lies in the word *simultaneously*: one basis, fixed once and for all, in which every group element takes block form at the same time. Finding that basis for representations of physical interest is, in a practical sense, what Lectures 5 and 6 are about.

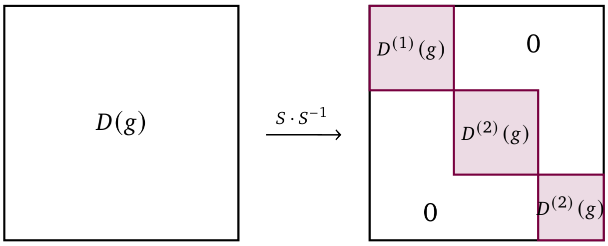

Figure 1.1: Maschke's theorem in matrix form: a single change of basis S brings every $D(g)$, for all $g \in G$ at once, to block-diagonal form with irreducible blocks. Blocks may repeat: here the irreducible representation $D^{(2)}$ occurs with multiplicity two.

Remark 1.22 (Notation, and what remains to prove)**.** Collecting equivalent blocks, we write

$$V \;\cong\; \bigoplus_i m_i\, V^{(i)}, \qquad \rho \;\cong\; \bigoplus_i m_i\, \rho^{(i)},$$

where $V^{(i)}$ runs over the inequivalent irreducible representations of G and the integer $m_i \geq 0$ is the *multiplicity* of $V^{(i)}$ in V. We reserve $n_i = \dim V^{(i)}$ for the irreducible dimension, so the total-dimension check is $\dim V = \sum_i m_i n_i$. Two caveats keep us honest. First, the decomposition is into *subspaces that are not unique* (already two copies of the trivial representation can be resliced into different pairs of invariant lines); what will turn out to be unique is the multiset of blocks — the multiplicities m_i. Second, nothing yet tells us how many irreducibles a given group has, or how to find the m_i without hunting for invariant subspaces by hand. Lectures 4 and 5 answer all of this with startling precision: G has finitely many irreducibles, as many as conjugacy classes, of dimensions satisfying $\sum_i (\dim V^{(i)})^2 = |G|$, and the multiplicities are computed by a one-line character formula.

Remark 1.23 (What the hypotheses buy). Both hypotheses of Maschke's theorem earn their keep. Drop finiteness of the group and Example 1.16 breaks the conclusion. Drop the field assumption and it breaks too: over fields of prime characteristic p dividing $|G|$, the averaging factor $1/|G|$ does not exist and complete reducibility fails (the subject called modular representation theory lives in the wreckage). Over $\mathbb{R}$, by contrast, the proof goes through verbatim with real inner products — though, as Example 1.13 showed, the *irreducible pieces* over $\mathbb{R}$ and over $\mathbb{C}$ can differ. In Part II, compactness of the Lie group will play the role finiteness plays here.

1.7 Worked example: the permutation representation of S_3

We now execute the whole programme by hand on the three-dimensional permutation representation of S_3 (Example 1.7), with its invariant subspaces $L = \mathbb{C}u_0$ and $V_0 = \{\sum_i v_i = 0\}$ from Example 1.15. Since $(1\,2)$ and $(1\,2\,3)$ generate S_3, every identity we need can be checked on these two elements alone.

Step 1: adapt the basis. Choose

$$u_0 = e_1 + e_2 + e_3, \qquad f_1 = e_1 - e_2, \qquad f_2 = e_2 - e_3,$$

so that (u_0, f_1, f_2) is a basis of $\mathbb{C}^3$ with u_0 spanning L and (f_1, f_2) spanning V_0.

Step 2: compute the blocks. The transposition $(1\,2)$ swaps e_1, e_2 and fixes e_3, so

$$f_1 \mapsto e_2 - e_1 = -f_1, \qquad f_2 \mapsto e_1 - e_3 = f_1 + f_2;$$

the cycle $(1\,2\,3)$ sends $e_1 \mapsto e_2 \mapsto e_3 \mapsto e_1$, so

$$f_1 \mapsto e_2 - e_3 = f_2, \qquad f_2 \mapsto e_3 - e_1 = -f_1 - f_2.$$

Reading off columns, the matrices of the subrepresentation on V_0 in the basis (f_1, f_2) are

$$B\big((1\,2)\big) = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B\big((1\,2\,3)\big) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix},$$

and indeed $B((1\,2))^2 = \mathbb{1}$, $B((1\,2\,3))^3 = \mathbb{1}$. In the full basis (u_0, f_1, f_2), with $S = \big(u_0 \,\big|\, f_1 \,\big|\, f_2\big)$ the change-of-basis matrix, we obtain in block form

$$S^{-1} D(g)\, S = \begin{pmatrix} 1 & 0 \\ 0 & B(g) \end{pmatrix} \qquad \text{for all } g \in S_3$$

— block-diagonal form achieved, with blocks of sizes 1 and 2, as direct multiplication (or the accompanying notebook) confirms.

Step 3: the 2×2 block is irreducible. A proper nonzero invariant subspace of V_0 would be a line, spanned by a common eigenvector of all the $B(g)$ — in particular of both generators. The eigenvectors of $B((1\,2))$ (eigenvalues -1 and $+1$) are the multiples of

$$w_- = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad w_+ = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

and neither line survives the cycle:

$$B\big((1\,2\,3)\big) w_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \nparallel w_-, \qquad B\big((1\,2\,3)\big) w_+ = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \nparallel w_+.$$

No common eigenvector, no invariant line: the standard representation of S_3 is irreducible. Altogether

$$\mathbb{C}^3 \;\cong\; V^{\mathrm{triv}} \oplus V^{\mathrm{std}},$$

the trivial representation plus a two-dimensional irreducible one, each with multiplicity one — and the third irreducible representation of S_3, the sign, does not appear at all. (That these multiplicities $1, 1, 0$ can be read off in one line, with no basis-hunting, is the promise of Lecture 5.)

Remark 1.24 (The geometry of the standard representation)**.** The decomposition has a picture. S_3 permutes the coordinate axes of $\mathbb{R}^3$; the diagonal direction u_0 is fixed, and the action turns the perpendicular plane V_0 into itself. The orthogonal projections of e_1, e_2, e_3 onto that plane form an equilateral triangle, which the group permutes — so the standard representation is nothing but the defining representation of $D_3 \cong S_3$ (Lecture 1) in disguise: the abstract block B and the geometric matrices of Example 1.5 are equivalent. One group, one irreducible two-dimensional representation, two coordinate descriptions.

Remark 1.25 (Computer algebra)**.** All of this mechanises cleanly. The permutation matrices are assembled as `IdentityMatrix[3][[PermutationList[g, 3]]]` for each group element `g` of `SymmetricGroup[3]`; the homomorphism property $D(g)D(h) = D(gh)$ is a one-line check over all pairs (mind Mathematica's left-to-right composition convention, Lecture 1). Conjugating by the explicit S of Step 2 displays the $1 \oplus 2$ block structure simultaneously for all six matrices, and `Eigenvectors` applied to the two generator blocks confirms the absence of a common eigenvector. The same notebook verifies Example 1.19 by computing the averaged Gram matrix M directly.

Summary.

- *Representation.* A homomorphism $G \to \mathrm{GL}(V)$, equivalently a linear action of G on V (Definition 1.1).
- *Function space.* G acts on functions by $(\Pi(g)\psi)(x) = \psi(g^{-1}x)$ (Theorem 1.9), with a dual representation on the dual space (Proposition 1.11).
- *Irreducibility.* An invariant subspace is one preserved by G; a representation is irreducible when it has no proper invariant subspace (Definition 1.14).
- *Unitarity.* Every representation of a finite group is unitarisable by averaging an inner product (Theorem 1.17).
- *Maschke.* Every finite-dimensional representation of a finite group is a direct sum of irreducibles (Theorem 1.21).

Tutorial. Exercise Sheet 3.

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